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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech. (EE) PT (Sem.–4)

POWER PLANT ENGINEERING

Subject Code : BTEE-406

M.Code : 72449

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

Answer briefly :

1. Discuss the function of a steam nozzle.

2. How the efficiency of a steam turbine can be calculated?

3. Discuss the role of a ash handling plant in a thermal power station.

4. What do you mean by hydrological cycle?

5. Define binding energy.

6. Define mass defect.

7. What are the various elements of a gas turbine plant?

8. Give any two differences between two stroke and four stroke diesel engines.

9. What are the advantages of combined operation of different power plants?

10 What is the function of an electrostatic precipitator in a power plant?

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SECTION-B

11. Draw a typical layout of a steam power plant. Explain the main features of layout?

12. What is a hydrograph? How can a flow duration curve be obtained from a hydrograph?

13. With the help of a suitable diagram explain the working of a Pressurized Water Reactor (PWR).

14. Differentiate between Open and Closed gas turbine cycles. Discuss the methods for improving the efficiency of gas turbine plants.

15. What are the advantages of diesel plants over thermal plants? Why diesel plants are not used for high capacity?

16. Explain briefly :

    (a) Combined operation of power plants

    (b) Pollution from nuclear and thermal power plants

17. A hydro-electric station has to operate with a mean head of 30 metres and is supplied from a reservoir lake at the rate of 6.93 m^3 per second. Calculate the power generated in kW. Assume density of water 1,000kg/m^3, load factor of station is 80%, mechanical efficiency of water turbine 90%. Neglect head loss in pipes, penstocks etc.

18 (a) Show that a mass defect of 1 amu is equivalent to about 931 MeVof energy.

    (b) Draw the layout of a diesel power plant and explain it in detail.

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